

Semantic core

What is true, under what conditions?

Stable claim

With negligible air resistance and constant g , vertical velocity changes uniformly: $v_y = u_y + a_y t$; with upward positive, $a_y = -g$.

Apex boundary

At the apex $v_y = 0$ but $a_y = -g$; horizontal velocity may remain nonzero.

Authority boundary

Core1 does not choose pedagogy, infer mastery, or authorize transfer.

Representation invariants

Same meaning across forms.

Story

Gravity changes v_y by the same signed amount each second.

Table/graph

Equal time steps give equal Δv_y ; the v_y -t graph is a straight line whose slope is a_y .

Misconceptions

Downward acceleration does not mean downward motion; a v_y -t graph is not the trajectory.

Core1 handoff

Stable semantics only.

Downstream

Core1A teaches; Core1B gathers learner evidence; Core2 describes challenge structure; Core2A authorizes transfer; Core2B runs transfer.

Teaching target

What must be built and repaired?

Capability

Track v_y through flight; distinguish velocity/acceleration; recognise apex; connect story, table, graph and equation.

Difficulty

Learners confuse negative acceleration with downward motion and apex with zero acceleration.

Meaning before formula

Build the physical idea first.

Phenomenon

A projectile may move upward while acceleration points downward.

Bridge

story -> signed change -> table -> v_y -t graph -> $v_y = u_y + a_y t$.

Progression

Fade support deliberately.

Worked

$t=1\text{ s}$, $v_y=12$, $a_y=-10 \rightarrow u_y=22\text{ m/s}$.

Faded

$t=2\text{ s}$, $v_y=6$, $a_y=-9 \rightarrow$ learner completes rearrangement.

Independent

$t=3\text{ s}$, $v_y=3$, $a_y=-9 \rightarrow$ learner chooses model, solves, explains applicability and checks.

Core1A handoff

Teaching authority, not mastery.

Invariant

One clock; signed vertical velocity; constant downward acceleration; apex $v_y=0$.

Boundary

Publication never proves learner mastery.

Prediction

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +2 \text{ m/s}$, $a_y = -10 \text{ m/s}^2$. Predict v_y one second later and explain.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Acceleration tells change, not motion direction now.

H2 One second changes v_y by -10.

H3 Add +2 and -10.

CHECK AFTER ATTEMPT - PREDICTION

$v_y = -8 \text{ m/s}$.

The projectile is then moving downward.

Read the pattern

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

time 0,1,2,3 s; v_y +22,+12,+2,-8. Find a_y and when direction changes.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Compare equal one-second intervals.

H2 Velocity drops by the same amount each second.

H3 Use $a = \Delta v / \Delta t$; direction changes at $v_y = 0$.

CHECK AFTER ATTEMPT - READ THE PATTERN

$a_y = -10 \text{ m/s}^2$.

Direction changes between 2 and 3 s.

Worked reconstruction

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At $t=1$ s, $v_y=+12$, $a_y=-10$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Use one clock.

H2 Take upward positive.

H3 $12=u_y-10$.

CHECK AFTER ATTEMPT - WORKED RECONSTRUCTION

$u_y=22$ m/s upward.

Check: $22-10=12$.

Independent use

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At $t=3$ s, $v_y=+3$, $a_y=-9$. Find u_y , state applicability, check.

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Recognise the event-clock structure.

H2 Use signed constant acceleration.

H3 $3=u_y-27$.

CHECK AFTER ATTEMPT - INDEPENDENT USE

$u_y=30$ m/s upward.

Model applies because vertical acceleration is constant and time is from launch.

Apex integrity

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

At the highest point, what are v_y and a_y , and why does the projectile fall afterward?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Separate velocity from acceleration.

H2 Zero velocity at an instant does not imply zero acceleration.

H3 Gravity remains downward.

CHECK AFTER ATTEMPT - APEX INTEGRITY

At apex $v_y=0$ and $a_y=-g$.

Gravity drives v_y negative immediately afterward.

Challenge semantics

What hidden structure does the problem demand?

Anchor demand

Given v_y at a stated elapsed time, infer another event variable using one global clock and signed constant-acceleration reasoning.

Hidden structure

recognise family -> choose sign -> place event on clock -> select equation -> solve target -> verify.

Transfer envelope

Surface may change while structure stays.

Variations

Near transfer, reversed target, event identification, representation shift, multi-step bridge.

Demand dimensions

Structural distance, representation change, model discrimination, multi-step bridge, synthesis, competitive mixing.

Boundary

A scalar T0-T8 label cannot be the sole authorizer.

Core2 handoff

Challenge semantics only.

Contribution

Problem-family identity, first-step reference, hint semantics, solution/verification requirements and transfer dimensions.

Legal pool input

What transfer is permissible?

Receipt

Projectile vertical-event teaching complete; learner evidence unknown.

Rule

Teaching completion permits compilation only for supported families/purposes; it does not prove transfer.

Legal candidates

Generated transfer under exact custody.

Near

$v_y=6$ at $t=2$ s, $a_y=-9$ -> find u_y . Expected 24 m/s.

Reversed target

$u_y=30$, $v_y=3$, $a_y=-9$ -> find t . Expected 3 s.

Fail-closed boundary

No silent expansion.

Legal

Direct, near transfer, reversed target.

Not in pool

New representation, synthesis or competitive mixing without supporting authority/evidence.

Direct recognition

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +12$ at $t = 1$ s, $a_y = -10$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Place launch and event on one clock.

H2 Use signed vertical constant acceleration.

H3 $12 = u_y - 10$.

CHECK AFTER ATTEMPT - DIRECT RECOGNITION

$u_y = 22$ m/s upward.

This is direct recognition of the taught structure.

Near transfer

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$v_y = +6$ at $t = 2$ s, $a_y = -9$. Find u_y .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Do not invent a new model because numbers changed.

H2 Keep the same sign and clock.

H3 $6 = u_y - 18$.

CHECK AFTER ATTEMPT - NEAR TRANSFER

$u_y = 24$ m/s upward.

Surface numbers changed; hidden structure did not.

Reversed target

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$u_y=30$, later $v_y=+3$, $a_y=-9$. Find t .

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Target changed; physics did not.

H2 Use the same relation for a different unknown.

H3 $3=30-9t$.

CHECK AFTER ATTEMPT - REVERSED TARGET

$t=3$ s.

Same invariant, different target.

Model selection

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

Which needs this model: horizontal range; infer u_y from later v_y and t ; centripetal acceleration?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 Look for stated vertical velocity at stated time.

H2 This capability is signed one-dimensional vertical velocity under constant acceleration.

H3 The second option matches.

CHECK AFTER ATTEMPT - MODEL SELECTION

The vertical-event option is the match.
Core2B tests recognition and selection, not only algebra.

Exit robustness

Open-ended attempt -> optional H1/H2/H3 -> same-page worked check.

Open-ended question

$u_y=36$, $v_y=0$, $a_y=-9$. What event is this and when?

YOUR RESPONSE

SELF-GUIDED HELP - USE ONLY IF NEEDED

H1 What event has $v_y=0$?

H2 Use the same one-clock signed model.

H3 $0=36-9t$.

CHECK AFTER ATTEMPT - EXIT ROBUSTNESS

Apex event; $t=4$ s.

Event recognition + model selection + changed target.

Six-layer distinction

Each layer answers a different question.

Core1

What is true?

Core1A

What must be taught?

Core1B

Can learner reconstruct/use it?

Core2

What challenge structure exists?

Core2A

What transfer is legal?

Core2B

Can learner recognise/select/transfer?

Authority leakage audit

Downstream cannot create upstream authority.

PASS

Core1A publication does not prove mastery; Core1B cannot create legality; Core2A does not prove mastery; Core2B cannot expand the legal pool.

Verdict

Process-golden validation.

PASS

Architecture is coherent and operational.

LIMIT

This does not prove real learner efficacy.

NEXT

Pilot with real responses; record hint use, failure dimension and repair success on a different transfer item.